

Math for Brain Data: Mathematical Foundation to Neurosystems

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Anastasia Simonoff, Freie Universitaet Berlin

Overview

1. Drawbacks of single-neuron models
2. From individual neurons to populations
3. Calculating average rates of neural populations
4. Population model intuition
5. Mean-field modeling
6. Excitatory and inhibitory populations
7. Wilson-Cowan E-I population model
8. Stability and equilibrium in population activity
9. Oscillations and rhythmic activity
10. Connectivity between populations
11. Emergent behaviour in neural systems

Intractability of Large Single-Neuron Models

- Single neuron models work for one neuron at a time
- Large number of neurons in brain:
 - ~86 billion neurons in the brain
 - $\sim 10^{14}$ - 10^{15} synapses
- Computationally intractable to simulate all at once
- Behavior and computation is done with multiple neurons working in tandem
 - Not individual neurons spiking
- Individual neurons are noisy and unreliable
 - Populations are more robust

Example: Going From Molecules to Gas Dynamics

- Example: Statistical Mechanics
- Do not track individual gas molecules
- Track temperature and pressure
 - Actions of many many gas molecules at once
 - Incredibly useful simplification
- $PV = nRT$ is an average over many molecules

Neuronal Populations

- What defines a population?
 - Neurons with similar properties
 - E.g., tuning curves, cell types, location
- Example population: V1 in the visual system
 - Early visual processing

Calculating Average Population Rates

- Multiple ways to calculate
- Temporal average
 - One neuron, #spikes / time
- Population average
 - Many neurons, #spiking at specific time
- Trial average
 - One neuron, averaged across trials
- Neuron average
 - Across many neurons
- Population activity
 - Fraction of neurons in the population firing in a small window

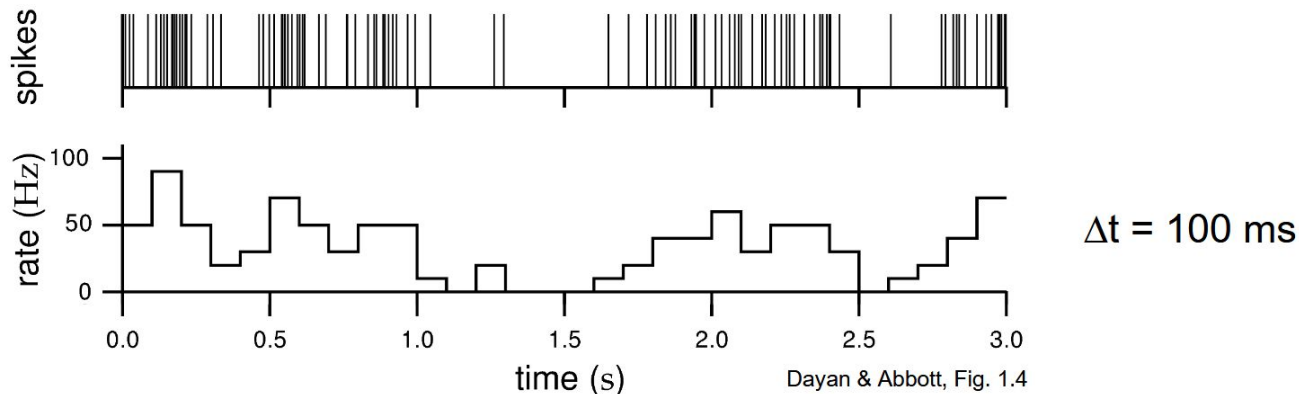
Spike Trains and Dirac Delta Functions

- Spike train:
 - $\{t_1, t_2, \dots, t_n\}$ with spike times $t_i, 0 \leq t_i \leq T$ for all i
 - T : duration of trial during which we record
- Assume spike train is a Dirac delta function
 - $\delta(\tau) = \infty$ if $\tau = 0$ else 0
- Define neural response function
 - $\rho(t) = \sum_{i=1}^n \delta(t - t_i)$



Temporal Average and Spike Count Rate

- Temporal average of ρ is a way to calculate the firing rate
- Done by binning (dividing the time into discrete intervals of duration Δt)
- Firing rate
$$r(k\Delta t) = \frac{n_k}{\Delta t} = \frac{1}{\Delta t} \int_{k\Delta t}^{(k+1)\Delta t} \rho(\tau) d\tau$$
- If $\Delta t = T$: spike count rate
$$r = \frac{1}{T} \int_0^T \rho(\tau) d\tau = \frac{n}{T}$$



Trial Average

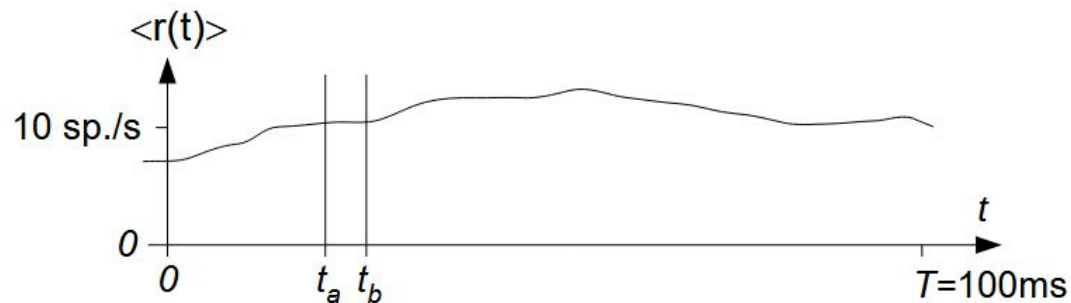
- Trial average: average of ρ over trials
- Trial-averaged response function: $\langle \rho(t) \rangle = \frac{1}{N} \sum_{k=1}^N \rho_k(t)$
- N is the total number of trials, k is the trial index
- $\langle \rho(t) \rangle$ is always a sum of δ functions. Not smooth
- $\langle \rho(t) \rangle$ is not useful on its own

Firing Rate and Average Firing Rate

- Population average: trial average + temporal average
- Firing rate (trial-averaged, time-dependent): $\langle r(t) \rangle = \frac{1}{\Delta t} \int_t^{t+\Delta t} \langle \rho(\tau) \rangle d\tau$
 - Much smaller Δt possible than single trial averages
- If $\Delta t = T$: average firing rate $\langle r \rangle = \frac{\langle n \rangle}{T} = \frac{1}{T} \int_0^T \langle \rho(\tau) \rangle d\tau$

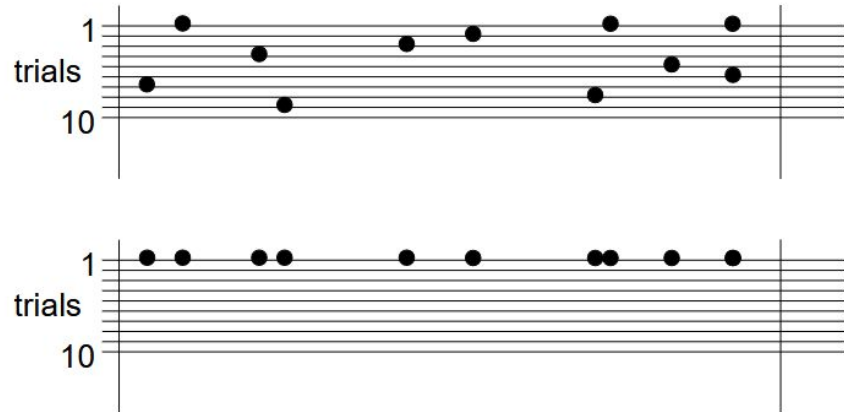
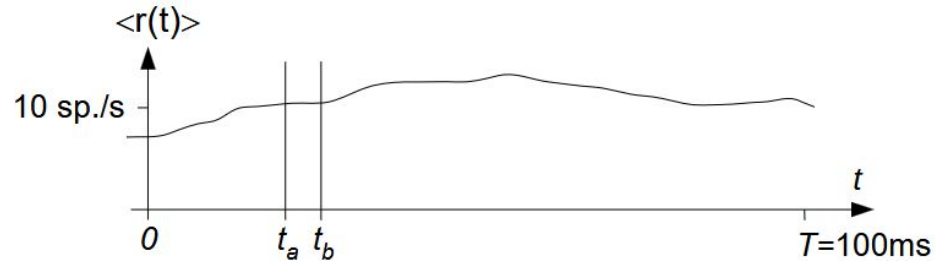
Interpreting Average Firing Rate

- How to interpret $\langle r(t) \rangle$?
- $T = 100sp * 10sp/s = 1sp/trial$
- Area under the curve:
 - Probability of observing a spike in a trial
 - Assuming area $\ll 1$



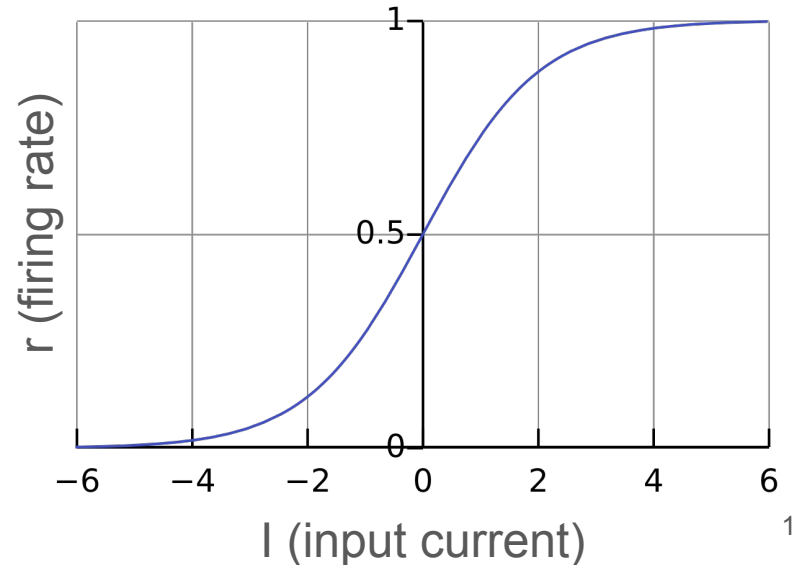
Dependent and Independent Firing

- Examples of single trials
- First: independent firing
- Second: interdependent firing



Population Model and Rate Equation

- Individual neurons have gain functions
- Generalize to populations
- Each activation function is stochastic -> firing curve is stochastic
- Assumes each neuron is identical
- Rate equation: $\tau \frac{dr}{dt} = -r + I_{in}$
 - τ : rate constant
 - I_{in} : input current
- Everything afterwards depends on this

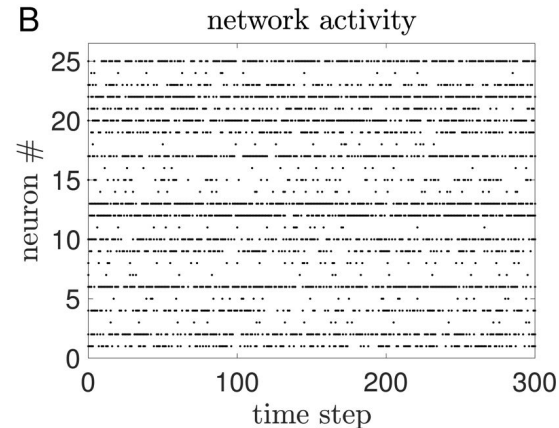
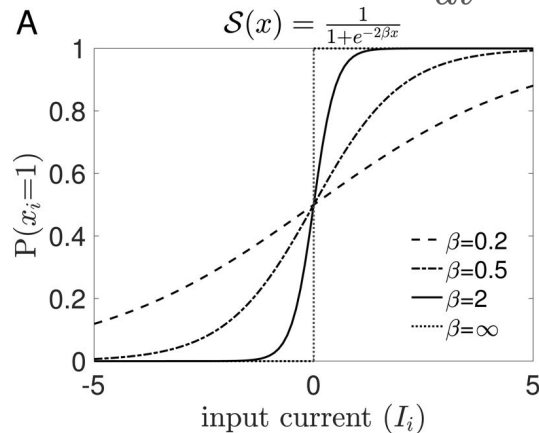


Mean-Field Modeling

- Purpose: average population activities from simulating individual neurons
 - Many brain signals are measured at larger scale than single neurons
 - E.g., local field potentials, recordings outside brain, etc.
- Want to get large-scale dynamics from small-scale interactions
 - Like temperature and pressure in physics
- Models must be self-consistent
 - Average neural input is the same as average population output
- Ignore correlations and finite-size fluctuations
 - Okay if N is large
 - Okay if correlations are weak

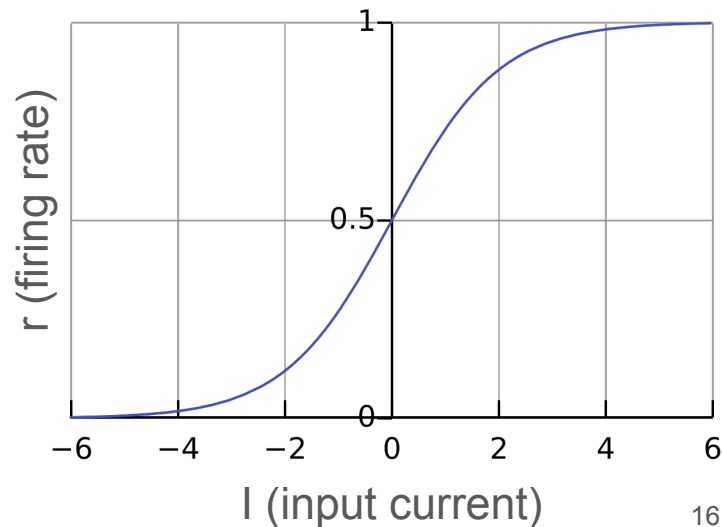
Mean-Field Modeling: Multiple Neurons

- Replace interaction between a neuron and its afferents with a mean field
 - Assume it receives a total input current equal to average of outputs of presynaptic neurons
- Helper function $S(x) = \frac{1}{1 + e^{-2\beta x}}$
 - Solution to the rate equation $\tau \frac{dr}{dt} = -r + I_{in}$



Mean-Field Modeling: Equations

- For one neuron:
- Rate equation: $\tau \frac{dr}{dt} = -r + I_{in}$
- Average firing rate of neuron i : $f_i = \frac{1}{1 + e^{-2\beta I_{i,in}}}$
 - Given an input current $I_{i,in}$
- Helper function S
 - $S(x) = \frac{1}{1 + e^{-2\beta x}}$
- Mean Field Equation:
 $f_i = S(\sum_{j \neq i} W_{ij} f_j + I_{i,in}), i = 1, \dots, N$
 - W_{ij} : synaptic strength between neurons i, j
- Set of N coupled equations
 - For firing rates of N neurons in the population



Mean-Field Modeling: Vector Form

- Mean Field Equation:

$$f_i = S(\sum_{j \neq i} W_{ij} f_j + I_{i,in}), i = 1, \dots, N$$

- Rewrite in vector notation:

$$\mathbf{f} = S(\mathbf{W}\mathbf{f} + \mathbf{I}_{in})$$

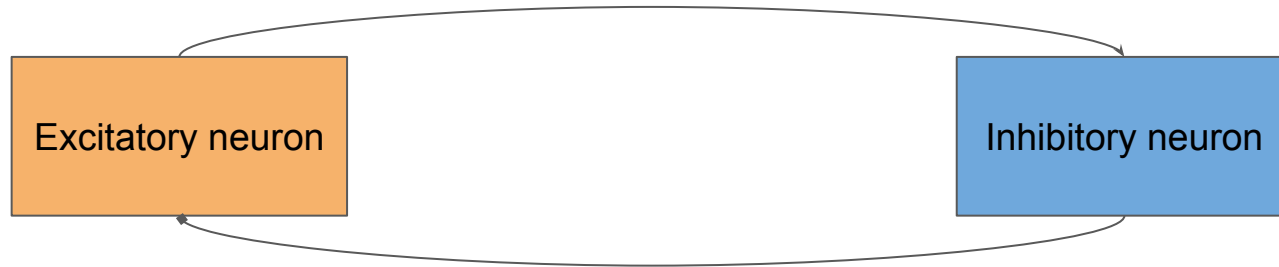
- $\mathbf{f}$: vector of firing rates
- $\mathbf{I}$: vector of input currents
- $\mathbf{W}$: matrix of synaptic connections

Excitatory and Inhibitory Neurons in the Brain

- How do we get interesting dynamics from populations?
- Need multiple kinds of subpopulations
 - Excitatory and inhibitory neurons
- Dale's law
 - Each neuron releases only one type of neurotransmitter
- Example neurotransmitters:
 - Glutamate: excitatory
 - Gamma-aminobutyric acid (GABA): inhibitory
 - Serotonin: inhibitory
 - Norepinephrine: excitatory or inhibitory, based on transmitter
 - Acetylcholine (ACh): excitatory in brain, inhibitory otherwise

Interactions of Excitatory and Inhibitory Neurons

- Need both excitatory and inhibitory neurons
 - Why not just one kind?
- Cortex: about 80% excitatory 20% inhibitory



Wilson-Cowan Model

- Idea: mathematically describe populations of E and I neurons
- Populations that are:
 - Large
 - Interconnected
 - Have sufficiently many synaptic connections
 - All receive same average external input

Wilson-Cowan Model: Equations

$$\tau_E \frac{dr_E}{dt} = -r_E + \sigma_E(w_{EE}r_E - w_{EI}r_I + I_{e,in})$$

$$\tau_I \frac{dr_I}{dt} = -r_I + \sigma_I(w_{IE}r_E - w_{II}r_I + I_{i,in})$$

- Written as differential equations
- r_E, r_I are the firing rates for the E, I populations
- τ_E, τ_I are the time constants
- w_{EI} is the connection strength from E to I
 - Same for EE, EI, IE, II
- σ_E, σ_I are the transfer functions (sigmoidal)
- $I_{e,in}, I_{i,in}$ are external input currents

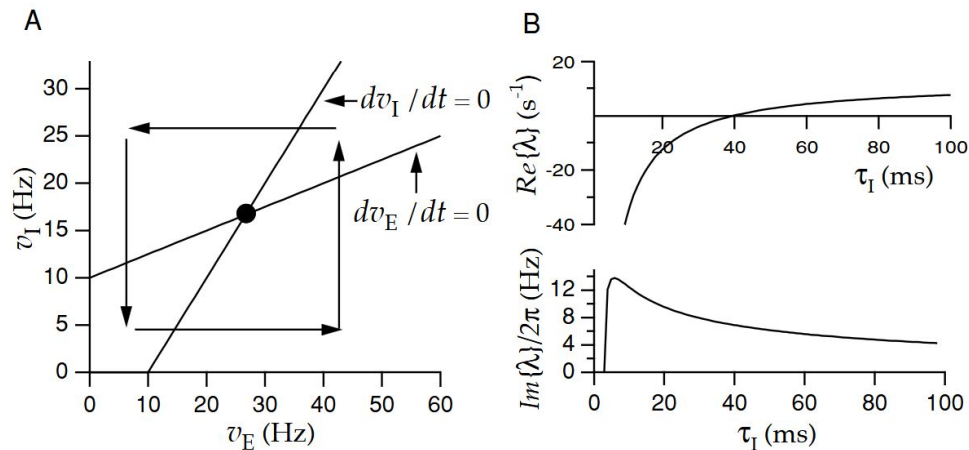
Nullclines and Trajectories

- Nullclines:

- $\frac{dv_I}{dt} = 0, \frac{dv_E}{dt} = 0$

- Trajectories:

- Movement of point in phase plane



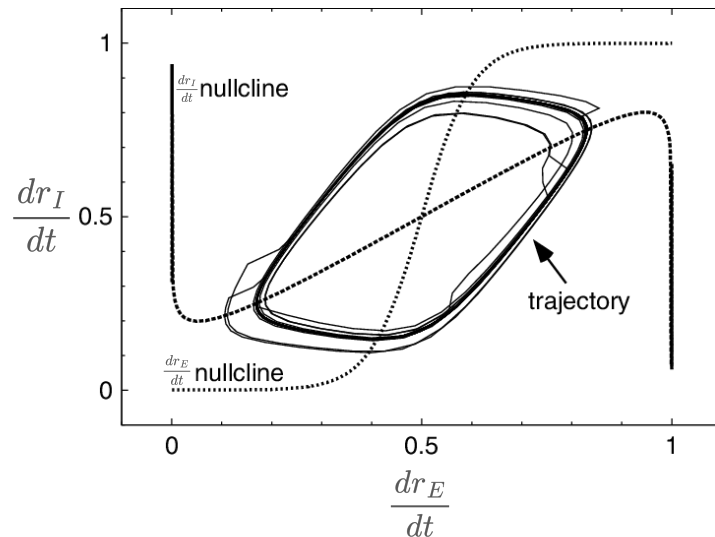
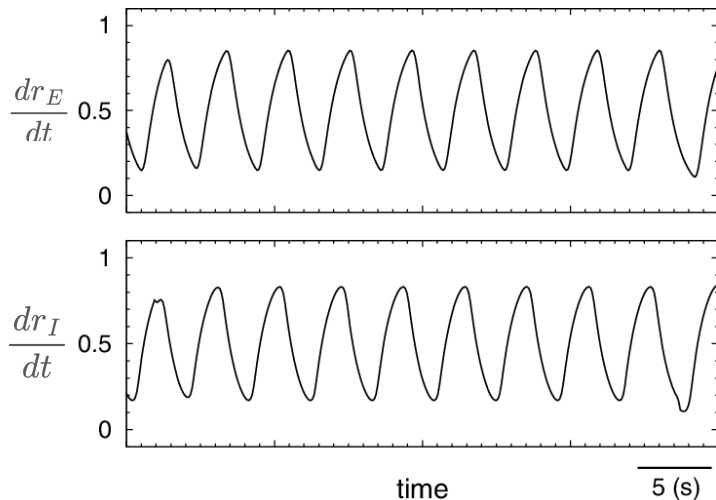
Oscillatory Activity and Trajectories

- Nullclines:

- $\frac{dr_E}{dt} = 0, \frac{dr_I}{dt} = 0$

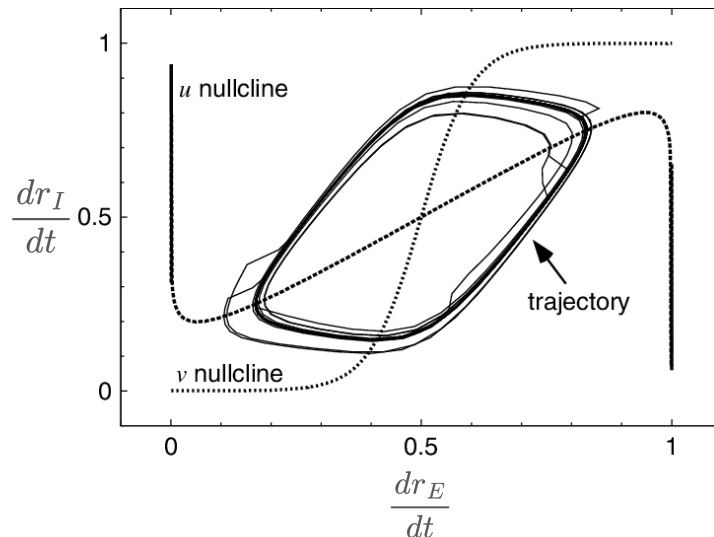
$$\tau_E \frac{dr_E}{dt} = -r_E + \sigma_E(w_{EE}r_E - w_{EI}r_I + I_{e,in})$$

$$\tau_I \frac{dr_I}{dt} = -r_I + \sigma_I(w_{IE}r_E - w_{II}r_I + I_{i,in})$$



Fixed Points and their Stability

- Fixed points:
 - $\frac{dr_E}{dt} = \frac{dr_I}{dt} = 0$
 - Only at intersection of nullclines
- Fixed point stability:
 - Evaluate Jacobian's eigenvalues at each fixed point
 - Jacobian: matrix of partials of each function
 - Calculate eigenvalues
 - If real part of both eigenvalues < 0 , stable
 - Otherwise, unstable
- Complexity:
 - If eigenvalues are real, exponential movement
 - If eigenvalues are complex, spirals/oscillations



$$J = \begin{pmatrix} \frac{\partial f_E}{\partial r_E} & \frac{\partial f_E}{\partial r_I} \\ \frac{\partial f_I}{\partial r_E} & \frac{\partial f_I}{\partial r_I} \end{pmatrix}$$

Eigenvectors and Eigenvalues

- Eigenvector $\mathbf{e}$ satisfies $\mathbf{J} \cdot \mathbf{e} = \lambda \mathbf{e}$
- Eigenvalue λ can be calculated by solving:
 - $\det(\mathbf{J} - \lambda \mathbf{I}) = 0$

$$J = \begin{pmatrix} \frac{\partial f_E}{\partial r_E} & \frac{\partial f_E}{\partial r_I} \\ \frac{\partial f_I}{\partial r_E} & \frac{\partial f_I}{\partial r_I} \end{pmatrix}$$

Evaluating the Jacobian

$$\frac{dr_E}{dt} = f_E = \frac{1}{\tau_E} \left[-r_E + \sigma_E(x_E) \right], \quad \frac{dr_I}{dt} = f_I = \frac{1}{\tau_I} \left[-r_I + \sigma_I(x_I) \right]$$

$$\text{where } x_E = w_{EE} r_E - w_{EI} r_I + I_{e,in}, \quad x_I = w_{IE} r_E - w_{II} r_I + I_{i,in}$$

Evaluate at fixed point (r_E^*, r_I^*) :

$$J = \begin{pmatrix} \frac{\partial f_E}{\partial r_E} & \frac{\partial f_E}{\partial r_I} \\ \frac{\partial f_I}{\partial r_E} & \frac{\partial f_I}{\partial r_I} \end{pmatrix} = \begin{pmatrix} \frac{1}{\tau_E} \left(-1 + w_{EE} \sigma'_E(x_E^*) \right) & \frac{1}{\tau_E} \left(-w_{EI} \sigma'_E(x_E^*) \right) \\ \frac{1}{\tau_I} \left(w_{IE} \sigma'_I(x_I^*) \right) & \frac{1}{\tau_I} \left(-1 - w_{II} \sigma'_I(x_I^*) \right) \end{pmatrix}$$

Identifying Stability

- For a 2x2 matrix (like the Jacobian):
$$\lambda_1 + \lambda_2 = \text{tr}(J)$$
$$\lambda_1 \lambda_2 = \det(J)$$

$$\text{tr}(J) = \frac{-1 + w_{EE} \sigma'_E(x_E^*)}{\tau_E} + \frac{-1 - w_{II} \sigma'_I(x_I^*)}{\tau_I}$$

$$\det(J) = \frac{1}{\tau_E \tau_I} \left[(-1 + w_{EE} \sigma'_E(x_E^*)) (-1 - w_{II} \sigma'_I(x_I^*)) + w_{EI} w_{IE} \sigma'_E(x_E^*) \sigma'_I(x_I^*) \right]$$

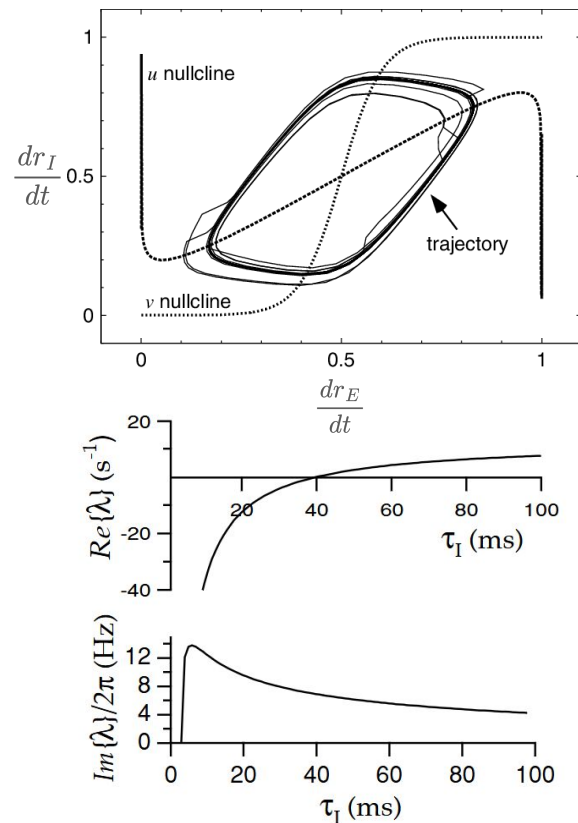
Stable: $\text{tr}(J) < 0$, $\det(J) > 0$.

Unstable: $\text{tr}(J) > 0$ AND/OR $\det(J) < 0$.

Hopf bifurcation: $\text{tr}(J) = 0$, $\det(J) > 0$.

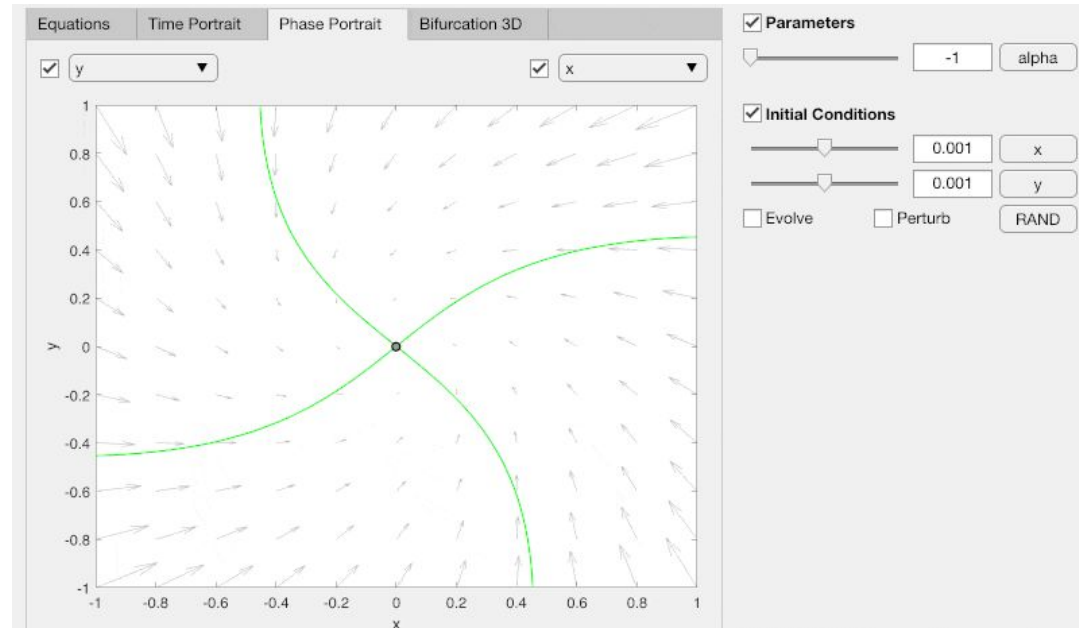
Oscillations and Rhythmic Activity

- Loop with E-I inhibition produces oscillations
 - Inhibitory feedback “overshoots”
- Limit cycles:
 - Sustained rhythms
- Arise from fixed points while varying I_{in}
 - Known as Hopf bifurcation
 - Where real part of complex eigenvalues changes sign



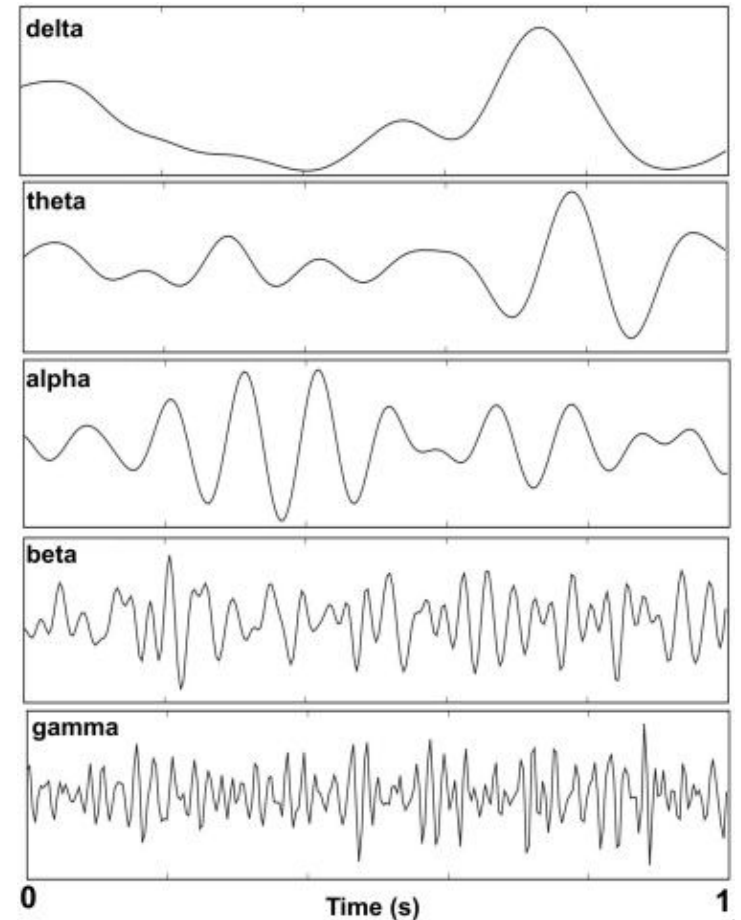
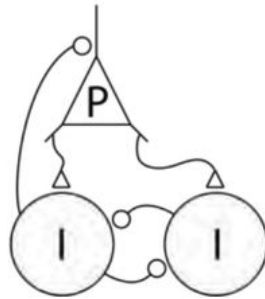
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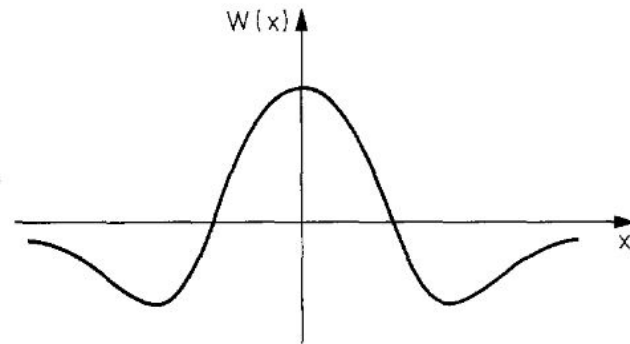
Oscillations and Rhythmic Activity

- EEG brain rhythms: oscillatory
 - At different time scales
- Example:
 - Gamma rhythm:
 - Pyramidal-Interneuron Network Gamma
 - Excitatory and Inhibitory cells



Connectivity Between Populations

- How do populations far apart interact?
 - Populations that are spatially distributed
- Structural connectivity:
 - How neurons are physically wired/connected
- Functional connectivity:
 - Causal relationships between neurons
- Neural fields:
 - How local stimulation affects neurons farther away
- Connection strength depends on distance
 - Represented with a kernel

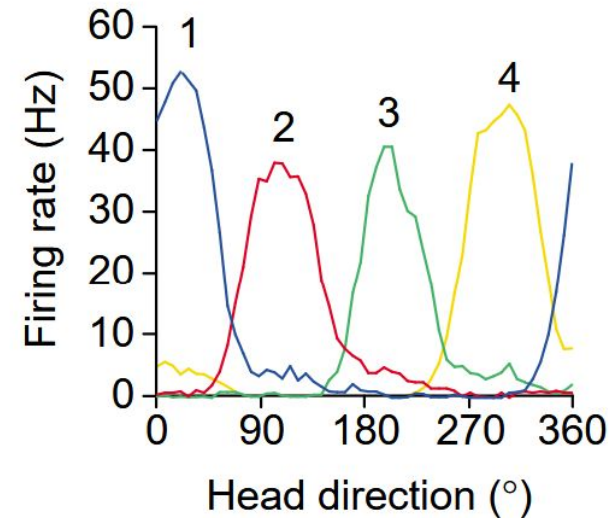
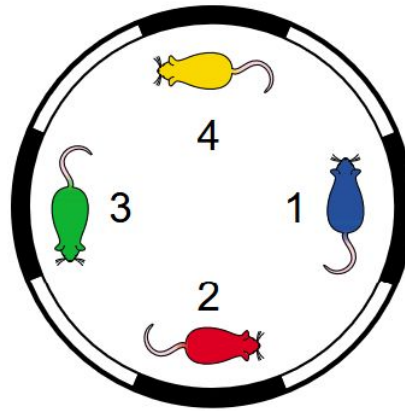


Emergent Behavior in Neural Systems

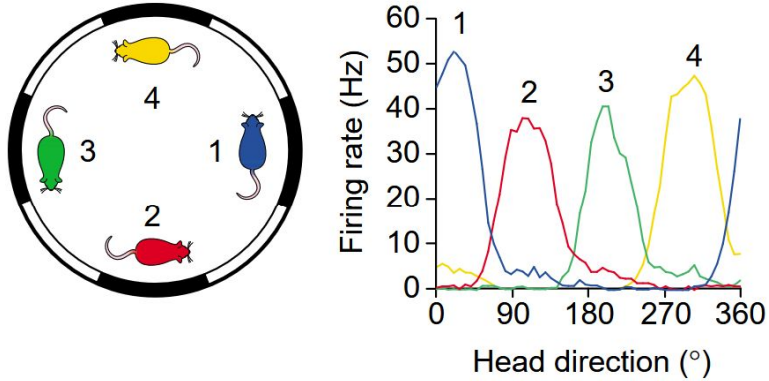
- Emergence:
 - Population properties not present in individual neurons
- Memories can be thought of as stable states
 - Similar to fixed points that we saw earlier
- Oscillations, such as brain waves
 - Emergent from neuronal populations

Emergent Behavior in Neural Systems

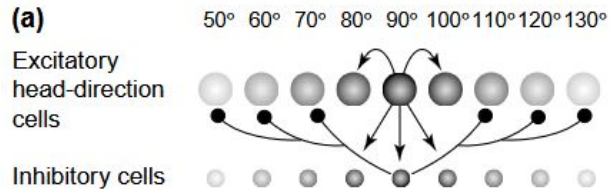
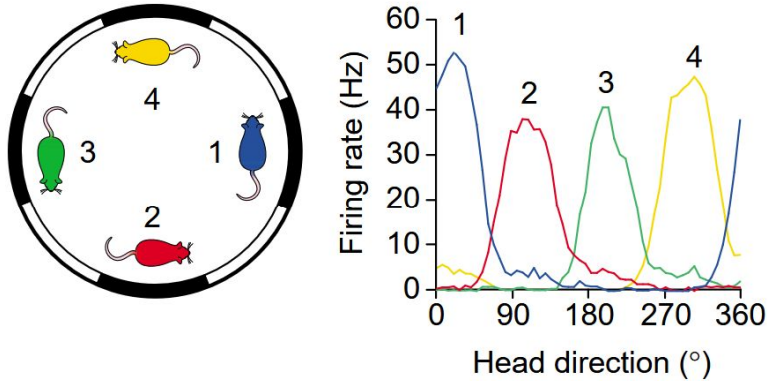
- Head-direction cells:
 - Fire when a rat's head is in its preferred direction
- Form an attractor network



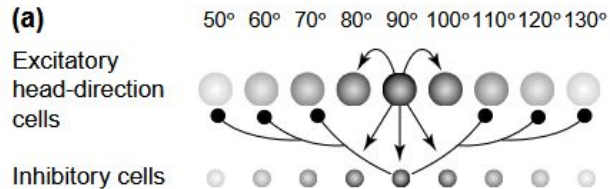
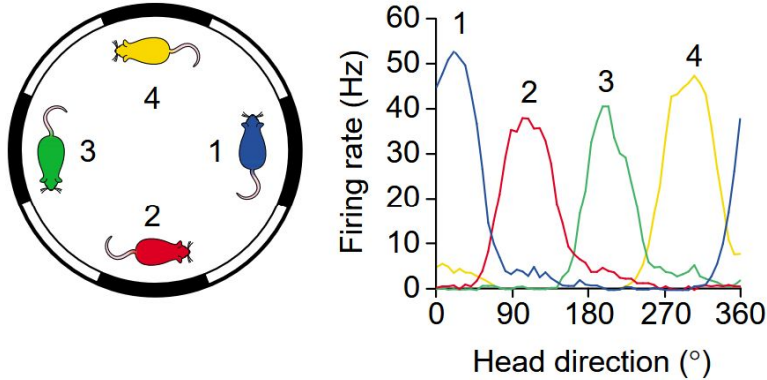
Emergent Behavior in Neural Systems



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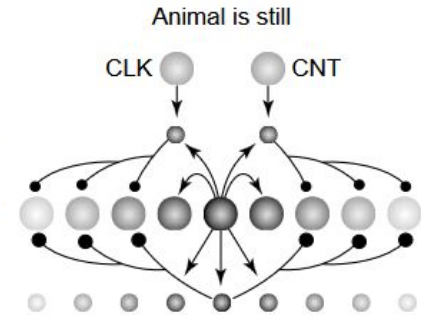
(b)

AV cells

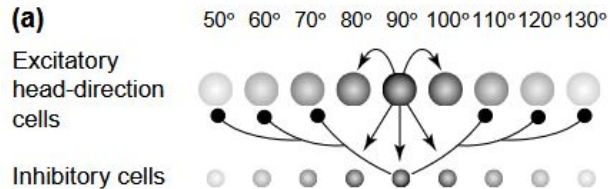
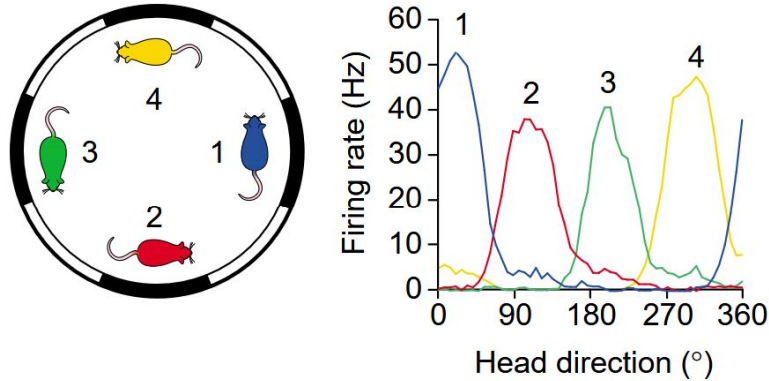
AV-by-HD cells

Excitatory head-direction cells

Inhibitory cells



Emergent Behavior in Neural Systems



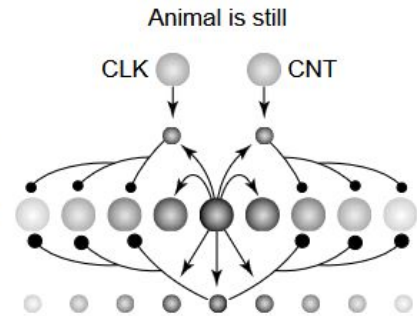
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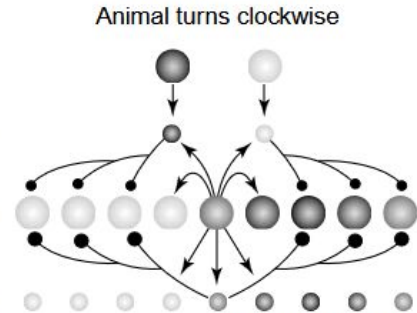


AV cells

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Questions?

Thank you!